Activity-2 : Multiple Inheritance

Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Program:

```
import java.util.Scanner;

public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        double c=sc.nextDouble();
        String code=sc.next();
        FoodOrder f=new FoodOrder(c,code);
        f.calculateDeliveryCharge();
        f.calculateDiscount();
        f.Display();
    }
}

interface DeliveryCharge{
    void calculateDeliveryCharge();
}

interface CouponDiscount{
    void calculateDiscount();
}

class FoodOrder implements DeliveryCharge,CouponDiscount{
    double foodcost;
    String couponcode;
    double deliverycharge;
    double Discount;

    FoodOrder(double foodcost,String couponcode){
        this.foodcost=foodcost;
        this.couponcode=couponcode;
    }

    public void calculateDeliveryCharge(){
        if(foodcost>=500)
            deliverycharge=0.0;
        else
            deliverycharge=50.0;
    }
```

```
    public void calculateDiscount(){
        if(couponcode.equals("SAVE10"))
            Discount=0.10*foodcost;
        else
            Discount=0.0;
    }

    public void Display(){
        double Final_Bill=foodcost+deliverycharge-Discount;
        System.out.println("Food Cost : " + foodcost);
        System.out.println("Delivery Charge : " + deliverycharge);
        System.out.println("Discount : " + Discount);
        System.out.println("Final Bill : " + Final_Bill);
    }
}
```

Input:

```
600
SAVE10
```

Output:

```
Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0
```

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Program:

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int nr=sc.nextInt();
        sc.nextLine();
        String gn=sc.nextLine();
        double rr=sc.nextDouble();
        HotelBill h=new HotelBill();
        h.get(nr,gn,rr);
        h.calculateServiceCharge();
        h.display();
    }
}
```

```
class Room{
   int room_num;
   String Guest_name;
   double room_rent;
   public void get(int room_num,String Guest_name,double room_rent){
      this.room_num=room_num;
      this.Guest_name=Guest_name;
      this.room_rent=room_rent;
   }
}
interface ServiceCharge{
   double calculateServiceCharge();
}

class HotelBill extends Room implements ServiceCharge{
   public double calculateServiceCharge(){
      if(room_rent>=5000)
         return room_rent*0.12;
      else
         return room_rent*0.08;
   }
   public void display(){
      double Service_charge=calculateServiceCharge();
      double final_bill=room_rent+Service_charge;
      System.out.println("Room Number : "+room_num);
      System.out.println("Guest Name : "+Guest_name);
      System.out.println("Room Rent : "+room_rent);
      System.out.println("Service Charge : "+Service_charge);
      System.out.println("Final Bill : "+final_bill);
   }
}
```

Input:

205
Ananya
6000

Output:

Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0